

Harmonisation Template for Cohort B

My Name

2025-03-10

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Preface

Here is the documentation of the data harmonisation step generated using [Quarto](https://quarto.org/docs/books). To learn more about Quarto books visit <https://quarto.org/docs/books>.

File Structure

Here is the file structure of the project used to generate the document.

```
harmonisation/                                # Root of the project template.
|
|   .quarto/ (not in repository)              # Folder to keep intermediate
files/folders                                # generated when Quarto renders
|                                              the files.
|
|   archive/                                  # Folder to keep previous books
and harmonised data.
|   |
|       reports/                             # Folder to keep previous versions
of
|   |   |
|   |   |   {some_date}_batch/              # Folder to keep {some_date}
documentation.                             version of
|   |   |   |                               # data harmonisation
|   |   |   |   Flowchart.xlsx              # Flowchart sheet to record
|   |   |   |   version control.
|   |   |   |
|   |   |   |   harmonised/                 # Folder to keep previous version
of harmonised data.
|   |   |   |   |
|   |   |   |   |   {some_date}_batch/     # Folder to keep {some_date}
version of
|   |   |   |   |   |
|   |   |   |   |   |   # harmonised data.
```

	Flowchart.xlsx	# Flowchart sheet to record
version control.		
codes/		# Folder to keep R/Quarto scripts
		# to run data harmonisation.
{cohort name}/		# Folder to keep Quarto scripts to
run		
		# data cleaning, harmonisation
		# and output them for each
cohort.		
	preprocessed_data/	# Folder to keep preprocessed
data.		
harmonisation_summary/		# Folder to keep Quarto scripts to
create		
		# data harmonisation summary
report.		
output/		# Folder to keep harmonised data.
cohort_harmonisation_script.R		# R script to render each {cohort
name}/ folder.		
		# folder into html, pdf and word
document.		
harmonisation_summary_script.R		# R script to render the
{harmonisation_summary}/		
		# folder into word document.
data-raw/		# Folder to keep cohort raw data
(.csv, .xlsx, etc.)		
{cohort name}/		# Folder to keep cohort raw data.
{data_dictionary}		# Data dictionary file that
correspond to the		
		# cohort raw data. Can be one
from the		
		# collaborator provide or
provided by us.		
Flowchart.xlsx		# Flowchart sheet to record
version control.		

	data-dictionary/	# Folder to keep data dictionary
		# used for harmonising data.
	Flowchart.xlsx	# Flowchart sheet to record
	version control.	
	data-input/	# Folder to keep data input file
		# for collaborators to fill in.
	Flowchart.xlsx	# Flowchart sheet to record
	version control.	
	docs/	# Folder to keep R functions
	documentation	
		# generated using
	pkgdown::build_site_external().	
	inst/	# Folder to keep arbitrary
	additional files	
		# to include in the project.
	WORDLIST	# File generated by
	spelling::update_wordlist()	
	man/	# Folder to keep R functions
	documentation	
		# generated using
	devtools::document().	
	{fun-demo}.Rd	# Documentation of the demo R
	function.	
	harmonisation-template.Rd	# High-level documentation.
	R/	# Folder to keep R functions.
	{fun-demo}.R	# Script with R functions.
	harmonisation-package.R	# Dummy R file for high-level
	documentation.	
	renv/ (not in repository)	# Folder to keep all packages
		# installed in the renv
	environment.	

reports/	# Folder to keep the most recent
data harmonisation	
	# documentation.
templates/	# Folder to keep template files
needed to generate	
	# data harmonisation
documentation efficiently.	
quarto-yaml/	# Folder to keep template files to
generate	
	# data harmonisation
documentation structure	
	# in Quarto.
quarto{cohort name}.yaml	# Quarto book template data
harmonisation documentation	
	# for {cohort name}.
_quarto_summary.yaml	# Quarto book template data
harmonisation summary.	
index-qmd/	# Folder to keep template files to
generate	
	# the preface page of the data
harmonisation	
	# documentation.
_index_report.qmd	# Preface template for each cohort
data harmonisation	
	# report.
_index_summary.qmd	# Preface template for data
harmonisation	
	# summary report.
tests/	# Folder to keep test unit files.
	# Files will be used by R package
testthat.	
.Rbuildignore	# List of files/folders to be
ignored while	
	# checking/installing the package.
.Renvron (not in repository)	
variables.	# File to set environment

.Rprofile (not in repository)	# R code to be run when R starts	
up.		
	# It is run after the .Renvi	
file is sourced.		
.Rhistory (not in repository)	# File containing R command	
history.		
.gitignore	# List of files/folders to be	
ignored while	# using the git workflow.	
.lintr	# Configuration for linting	
	# R projects and packages using	
linter.		
.renvignore	# List of files/folders to be	
ignored when	# renv is doing its snapshot.	
DESCRIPTION[*]	# Overall metadata of the project.	
LICENSE	# Content of the MIT license	
generated via	# usethis::use_mit_license().	
LICENSE.md	# Content of the MIT license	
generated via	# usethis::use_mit_license().	
NAMESPACE	# List of functions users can use	
or imported	# from other R packages. It is	
	# by devtools::document().	
generated		
README.md	# GitHub README markdown file	
generated by Quarto.		
README.qmd	# GitHub README quarto file used	
to generate README.md.		
_pkgdown.yml	# Configuration for R package	
documentation		


```

|                                     # using
pkgdown:::build_site_external().
|
|   _quarto.yml                       # Configuration for Quarto book
generation.
|                                     # It is also the project
configuration file.
|
|   csl_file.csl                     # Citation Style Language (CSL)
file to ensure
|                                     # citations follows the Lancet
journal.
|
|   custom-reference.docx            # Microsoft word template for data
harmonisation
|                                     # documentation to Word.
|
|   harmonisation_template.Rproj      # RStudio project file.
|
|   index.qmd                        # Preface page of Quarto book
content.
|
|   references.bib                   # Bibtex file for Quarto book.
|
|   renv.lock                         # Metadata of R packages installed
generated
|                                     # using renv::snapshot().

```

[*] These files are automatically created but user needs to manually add some information.

Part I

Cohort B Cleaning

1 R Package And Environment

1.1 R Packages Used

Here are the R packages used in this analysis.

```
harmonisation::get_r_package_info() |>  
  knitr::kable()
```

package	version	date	source
dplyr	1.1.4	2023-11-17	RSPM
fontawesome	0.5.3	2024-11-16	RSPM
forcats	1.0.1	2025-09-25	RSPM
glue	1.8.0	2024-09-30	RSPM
harmonisation	1.0.0.0	2025-11-12	local
here	1.0.2	2025-09-15	RSPM
htmltools	0.5.8.9000	2024-06-07	https://j~
magrittr	2.0.4	2025-09-12	RSPM
nanoparquet	0.4.2	2025-02-22	RSPM
openxlsx	4.2.8.1	2025-10-31	RSPM
pointblank	0.12.2	2024-10-23	RSPM
purrr	1.2.0	2025-11-04	RSPM
quarto	1.5.1	2025-09-04	RSPM
reactable	0.4.4	2023-03-12	RSPM
sessioninfo	1.2.3	2025-02-05	RSPM
stringr	1.6.0	2025-11-04	RSPM
testthat	3.2.3	2025-01-13	RSPM
tibble	3.3.0	2025-06-08	RSPM
tidyr	1.3.1	2024-01-24	RSPM
vroom	1.6.6	2025-09-19	RSPM

1.2 R Platform Information

Here are the R platform environment used in this analysis.

```
harmonisation::get_r_platform_info() |>
  knitr::kable()
```

setting	value
version	R version 4.5.1 (2025-06-13 ucrt)
os	Windows 11 x64 (build 26200)
system	x86_64, mingw32
ui	RTerm
language	(EN)
collate	English_Singapore.utf8
ctype	English_Singapore.utf8
tz	Asia/Singapore
date	2025-11-12
pandoc	3.6.3 @ C:/Program Files/RStudio/resources/app/bin/quarto/bin/tools/ (via rmarkdown)
quarto	1.8.21 @ C:/Program Files/Quarto/bin/quarto.exe/ (via quarto)
knitr	1.50 from RSPM

1.3 Data dictionary

Check to see if the data dictionary 20250310_data_dictionary.xlsx exists.

```
dict_relative_path <- fs::path(
  "data-raw",
  "data_dictionary",
  params$data_dictionary
)

dict_path <- here::here(dict_relative_path)

if (!file.exists(dict_path)) {
  stop(glue::glue("Input data dictionary {dict_path} cannot be found"))
}
```

2 Read Cohort B Data

2.1 Read Data

We read the file `data_to_harmonise_age_issue.csv` using `vroom::vroom`

```
cohort_B_data <- vroom::vroom(  
  file = here::here("data-raw",  
                    "Cohort_B",  
                    "data_to_harmonise_age_issue.csv"),  
  delim = ",",  
  col_select = 1:2,  
  show_col_types = FALSE,  
  col_types = list(  
    ID = vroom::col_character(),  
    Age = vroom::col_integer()  
  )  
) |>  
dplyr::rename(cohort_unique_id = "ID") |>  
# Remove rows when the ID value is NA  
dplyr::filter(!is.na(.data[["cohort_unique_id"]])) |>  
# Remove white spaces in column names  
dplyr::rename_all(stringr::str_trim) |>  
# Check if cohort id is unique  
pointblank::rows_distinct(  
  columns = "cohort_unique_id",  
)
```

To safeguard a csv file with issues, we can use the function `vroom::problems`

If there are issues with the data, the output of `vroom::problems` will be a `tibble`.

```
cohort_B_data |>  
  vroom::problems()
```

```
# A tibble: 3 x 5  
  row    col expected    actual  file  
  <int> <int> <chr>      <chr>  <chr>
```

```

1      4      2 an integer missing
D:/Jeremy/PortableR/RPortableWorkDirectory/har~
2     10      2 an integer missing
D:/Jeremy/PortableR/RPortableWorkDirectory/har~
3     17      2 an integer missing
D:/Jeremy/PortableR/RPortableWorkDirectory/har~

```

To check for this in an automatically, we can use `pointblank::expect_row_count_match`

```

cohort_B_data |>
  vroom::problems() |>
  pointblank::expect_row_count_match(count = 0)

```

Error: Row counts for the two tables did not match.
 The `expect\_row\_count\_match()` validation failed beyond the absolute threshold level (1).
 * failure level (1) >= failure threshold (1)

Suppose we have a csv file with no issues, we can safeguard it with the following code.

```

cohort_B_data <- vroom::vroom(
  file = here::here("data-raw",
                    "Cohort_B",
                    "data_to_harmonise.csv"),
  delim = ",",
  col_select = 1:8,
  show_col_types = FALSE,
  col_types = list(
    ID = vroom::col_character(),
    Age = vroom::col_integer(),
    Sex = vroom::col_character(),
    Height = vroom::col_double(),
    Weight = vroom::col_double(),
    `Smoke History` = vroom::col_character(),
    `Chest Pain Character` = vroom::col_character(),
    Dyspnea = vroom::col_character()
  )
) |>
dplyr::rename(cohort_unique_id = "ID") |>
# Remove rows when the ID value is NA
dplyr::filter(!is.na(.data[["cohort_unique_id"]])) |>
# Remove white spaces in column names
dplyr::rename_all(stringr::str_trim) |>
# Check if cohort id is unique
pointblank::rows_distinct(
  columns = "cohort_unique_id",

```

```
)  
  
cohort_B_data |>  
  vroom::problems() |>  
  pointblank::expect_row_count_match(count = 0)
```

2.2 Write Preprocessed File

We output data to be used for the next session.

```
cohort_B_data |>  
  fst::write_fst(  
    path = here::here(params$analysis_folder,  
                      params$harmonisation_folder,  
                      params$preprocessing_folder,  
                      "01_Cohort_B_cleaned.fst")  
  )
```

3 Extract Demographic

3.1 Read Preprocessed File

We read output data from the previous section.

3.2 Demographics and Behavioral parameters

3.2.1 Age and Sex

`age_years` will be mapped from the column `Age`.

`sex` is grouped as follows:

Sex	sex
Female	0
Male	1

```
age_gender_data <- cohort_B_data |>
  dplyr::select(c("cohort_unique_id",
                  "Age",
                  "Sex")) |>
  pointblank::col_vals_expr(
    expr = ~ harmonisation::is_integer_vector(
      cohort_A_data[["age"]],
      allow_na = TRUE)
  ) |>
  dplyr::mutate(
    # Convert age to type integer
    age_years = as.integer(.data[["Age"]]),
    # Convert categorical columns to factors
    sex = dplyr::case_when(
      .data[["Sex"]] == "Female" ~ "0",
      .data[["Sex"]] == "Male" ~ "1",
      .default = NA_character_
    ),
    `Sex` = forcats::fct_relevel(
```



```

    .data[["Sex"]],
    c("Female", "Male")
  ),
  sex = forcats::fct_relevel(
    .data[["sex"]],
    c("0", "1")),
) |>
dplyr::relocate(
  "sex",
  .after = "Sex"
) |>
dplyr::relocate(
  "age_years",
  .after = "Age"
) |>
pointblank::col_vals_in_set(
  columns = "sex",
  set = c("0", "1")
) |>
pointblank::col_vals_between(
  columns = "age_years",
  left = 0,
  right = 100,
  inclusive = c(FALSE, TRUE),
  na_pass = TRUE
)

```

```

if (params$show_table) {
  age_gender_data |>
    dplyr::distinct(.data[["Sex"]],
                    .keep_all = TRUE) |>
    knitr::kable()
}

```

cohort_unique_id	Age	age_years	Sex	sex
B001	32	32	Female	0
B003	80	80	Male	1

Remove unnecessary columns so that we can merge with the other fields.

```

age_gender_data <- age_gender_data |>
  dplyr::select(-c("Age", "Sex"))

```

3.2.2 Height, Weight, BMI and BSA

height_cm will be mapped from the column Height. weight_kg will be mapped from the column Weight.

bsa_m2 in m^2 will be calculated as $\sqrt{(\text{Height(cm)} \times \text{Weight(kg)})/3600}$ bmi will be calculated as $\text{Weight(kg)} / ((\text{Height(m)})^2)$

All values are then converted to two decimal places.

```
body_measurement_data <- cohort_B_data |>
  dplyr::select(c("cohort_unique_id",
                  "Height", "Weight")) |>
  dplyr::mutate(
    height_cm = .data[["Height"]],
    weight_kg = .data[["Weight"]],
    bsa_m2 = sqrt((.data[["height_cm"]] * .data[["weight_kg"]]) / 3600),
    bsa_m2 = harmonisation::round_to_nearest_digit(.data[["bsa_m2"]],
    ↪ digits = 2),
    bmi = .data[["weight_kg"]] / ((.data[["height_cm"]] / 100)^2),
    bmi = harmonisation::round_to_nearest_digit(.data[["bmi"]], digits =
    ↪ 2),
    height_cm = harmonisation::round_to_nearest_digit(.data[["height_cm"]],
    ↪ digits = 2),
    weight_kg = harmonisation::round_to_nearest_digit(.data[["weight_kg"]],
    ↪ digits = 2)
  ) |>
  pointblank::col_vals_between(
    columns = "bmi",
    left = 10,
    right = 50,
    inclusive = c(TRUE, TRUE),
    na_pass = TRUE
  )
```

```
if (params$show_table) {
  body_measurement_data |>
    head(n = 5) |>
    knitr::kable()
}
```

cohort_unique_id	Height	Weight	height_cm	weight_kg	bsa_m2	bmi
B001	170	63	170	63	1.72	21.80
B002	167	71	167	71	1.81	25.46
B003	184	77	184	77	1.98	22.74

cohort_unique_id	Height	Weight	height_cm	weight_kg	bsa_m2	bmi
B004	160	83	160	83	1.92	32.42
B005	155	61	155	61	1.62	25.39

Remove unnecessary columns so that we can merge with the other fields.

```
body_measurement_data <- body_measurement_data |>
  dplyr::select(-c("Height", "Weight"))
```

3.2.3 Smoking History

`smoke_current` is grouped as follows:

Smoke History	smoke_current
non-smoker	0
past smoker	0
current smoker	1
NA	-1

`smoke_past` is grouped as follows:

Smoke History	smoke_past
non-smoker	0
past smoker	1
current smoker	0
NA	-1

We do a check to ensure that we can only have these scenarios

- `smoke_current` as 1 and `smoke_past` as 0 for current smokers
- `smoke_current` as 0 and `smoke_past` as 1 for past smokers
- `smoke_current` as 0 and `smoke_past` as 0 for non-smokers
- `smoke_current` as -1 and `smoke_past` as -1 for unknown

```
smoking_data <- cohort_B_data |>
  dplyr::select(c("cohort_unique_id",
                  "Smoke History")) |>
  dplyr::mutate(
    smoke_current = dplyr::case_when(
      is.na(.data[["Smoke History"]]) ~ "-1",
```

```

      .data[["Smoke History"]] == "non-smoker" ~ "0",
      .data[["Smoke History"]] == "past smoker" ~ "0",
      .data[["Smoke History"]] == "current smoker" ~ "1",
      .default = NA_character_
    ),
    smoke_current = forcats::fct_relevel(
      .data[["smoke_current"]],
      c("0", "1")),
    smoke_past = dplyr::case_when(
      is.na(.data[["Smoke History"]]) ~ "-1",
      .data[["Smoke History"]] == "non-smoker" ~ "0",
      .data[["Smoke History"]] == "past smoker" ~ "1",
      .data[["Smoke History"]] == "current smoker" ~ "0",
      .default = NA_character_
    ),
    smoke_past = forcats::fct_relevel(
      .data[["smoke_past"]],
      c("0", "1")),
    `Smoke History` = forcats::fct(
      .data[["Smoke History"]]
    )
  ) |>
pointblank::col_vals_in_set(
  columns = c("smoke_current", "smoke_past"),
  set = c("0", "1", "-1")
) |>
pointblank::col_vals_expr(
  expr = pointblank::expr(
    (.data[["smoke_current"]] == "1" & .data[["smoke_past"]] == "0") |
    (.data[["smoke_current"]] == "-1" & .data[["smoke_past"]] == "-1") |
    (.data[["smoke_current"]] == "0" & .data[["smoke_past"]] %in% c("0",
↵ "1"))
  )
)

```

```

if (params$show_table) {
  smoking_data |>
    dplyr::distinct(.data[["Smoke History"]],
                    .keep_all = TRUE) |>
    knitr::kable()
}

```

cohort_unique_id	Smoke History	smoke_current	smoke_past
B001	non-smoker	0	0
B002	current smoker	1	0
B004	past smoker	0	1
B017	NA	-1	-1

Remove unnecessary columns so that we can merge with the other fields.

```
smoking_data <- smoking_data |>
  dplyr::select(-c("Smoke History"))
```

3.2.4 Chest Pain

3.2.4.1 Shortness of Breath

have_sob is grouped as follows:

Dyspnea	have_sob
no	0
yes	1

```
shortness_of_breath_data <- cohort_B_data |>
  dplyr::select(c("cohort_unique_id", "Dyspnea")) |>
  dplyr::mutate(
    have_sob = dplyr::case_when(
      .data[["Dyspnea"]] == "no" ~ "0",
      .data[["Dyspnea"]] == "yes" ~ "1",
      .default = NA_character_
    ),
    have_sob = forcats::fct_relevel(
      as.character(.data[["have_sob"]]),
      c("0", "1")),
    Dyspnea = forcats::fct_relevel(
      as.character(.data[["Dyspnea"]]),
      c("no", "yes")),
  ) |>
  pointblank::col_vals_in_set(
    columns = c("have_sob"),
    set = c("0", "1", "-1")
  )
```

```

if (params$show_table) {
  shortness_of_breath_data |>
    dplyr::distinct(.data[["Dyspnea"]],
                    .keep_all = TRUE) |>
    knitr::kable()
}

```

cohort_unique_id	Dyspnea	have_sob
B001	no	0
B002	yes	1

Remove unnecessary columns so that we can merge with the other fields.

```

shortness_of_breath_data <- shortness_of_breath_data |>
  dplyr::select(-c("Dyspnea"))

```

3.2.4.2 Have chest pain or not

have_chest_pain is grouped as follows:

Chest Pain Character	have_chest_pain
no chest pain	0
typical, atypical or nonanginal	1

```

have_chest_pain_data <- cohort_B_data |>
  dplyr::select(c("cohort_unique_id", "Chest Pain Character")) |>
  dplyr::mutate(
    have_chest_pain = dplyr::case_when(
      .data[["Chest Pain Character"]] %in% c("no chest pain") ~ "0",
      .data[["Chest Pain Character"]] %in% c("typical", "atypical",
↵ "nonanginal") ~ "1",
      .default = NA_character_
    ),
    have_chest_pain = forcats::fct_relevel(
      .data[["have_chest_pain"]],
      c("0", "1")
    ),
    `Chest Pain Character` = forcats::fct_relevel(
      as.character(.data[["Chest Pain Character"]]),
      c("no chest pain", "typical", "atypical", "nonanginal")
    )
  )

```

```

)
) |>
pointblank::col_vals_in_set(
  columns = c("have_chest_pain"),
  set = c("0", "1")
)

```

```

if (params$show_table) {
  have_chest_pain_data |>
    dplyr::distinct(.data[["Chest Pain Character"]],
                    .keep_all = TRUE) |>
    knitr::kable()
}

```

cohort_unique_id	Chest Pain Character	have_chest_pain
B001	atypical	1
B002	no chest pain	0
B005	typical	1
B006	nonanginal	1

Remove unnecessary columns so that we can merge with the other fields.

```

have_chest_pain_data <- have_chest_pain_data |>
  dplyr::select(-c("Chest Pain Character"))

```

3.2.4.3 Symptomatic or Asymptomatic

symptoms is grouped as follows:

have_sob	have_chest_pain	symptoms
-1	-1	-1
0	0	0
0 or 1	1	1
1	0	2

```

symptoms_data <- cohort_B_data |>
  dplyr::select(c("cohort_unique_id")) |>
  dplyr::inner_join(shortness_of_breath_data,
                    by = dplyr::join_by("cohort_unique_id"),
                    unmatched = "error",

```

```

        relationship = "one-to-one") |>
dplyr::inner_join(have_chest_pain_data,
  by = dplyr::join_by("cohort_unique_id"),
  unmatched = "error",
  relationship = "one-to-one") |>
dplyr::mutate(
  symptoms = dplyr::case_when(
    (.data[["have_chest_pain"]] == "-1" &
     .data[["have_sob"]] == "-1"
    ) ~ "-1",
    (.data[["have_chest_pain"]] == "0" &
     .data[["have_sob"]] == "0"
    ) ~ "0",
    (.data[["have_chest_pain"]] == "1" &
     .data[["have_sob"]] %in% c("0", "1")
    ) ~ "1",
    (.data[["have_chest_pain"]] == "0" &
     .data[["have_sob"]] == "1"
    ) ~ "2",
    .default = NA_character_
  ),
  symptoms = forcats::fct_relevel(
    .data[["symptoms"]],
    c("0", "1", "2"))
) |>
pointblank::col_vals_in_set(
  columns = c("symptoms"),
  set = c("0", "1", "2")
)

```

```

if (params$show_table) {
  symptoms_data |>
    dplyr::distinct(.data[["have_chest_pain"]], .data[["have_sob"]],
      .keep_all = TRUE) |>
    knitr::kable()
}

```

cohort_unique_id	have_sob	have_chest_pain	symptoms
B001	0	1	1
B002	1	0	2
B003	1	1	1
B007	0	0	0

Remove unnecessary columns so that we can merge with the other fields.


```
symptoms_data <- symptoms_data |>
  dplyr::select(-c("have_chest_pain", "have_sob"))
```

3.2.4.4 Chest Pain Type

chest_pain_type is grouped as follows:

Dyspnea	Chest Pain Character	chest_pain_type
no	no chest pain	0
no or yes	typical	1
no or yes	atypical	2
no or yes	nonanginal	3
yes	no chest pain	4

```
chest_pain_type_data <- cohort_B_data |>
  dplyr::select(c("cohort_unique_id", "Chest Pain Character", "Dyspnea"))
  |>
  dplyr::mutate(
    chest_pain_type = dplyr::case_when(
      (.data[["Chest Pain Character"]] == "no chest pain" &
        .data[["Dyspnea"]] == "no"
      ) ~ "0",
      (.data[["Chest Pain Character"]] == "typical" &
        .data[["Dyspnea"]] %in% c("no", "yes")
      ) ~ "1",
      (.data[["Chest Pain Character"]] == "atypical" &
        .data[["Dyspnea"]] %in% c("no", "yes")
      ) ~ "2",
      (.data[["Chest Pain Character"]] == "nonanginal" &
        .data[["Dyspnea"]] %in% c("no", "yes")
      ) ~ "3",
      (.data[["Chest Pain Character"]] == "no chest pain" &
        .data[["Dyspnea"]] == "yes"
      ) ~ "4",
      .default = NA_character_
    ),
    `Chest Pain Character` = forcats::fct_relevel(
      as.character(.data[["Chest Pain Character"]]),
      c("no chest pain", "typical", "atypical", "nonanginal")
    ),
    `Dyspnea` = forcats::fct_relevel(
      as.character(.data[["Dyspnea"]]),
      c("no", "yes")
    )
  )
```

```

),
chest_pain_type = forcats::fct_relevel(
  .data[["chest_pain_type"]],
  c("0", "1", "2", "3"))
) |>
dplyr::relocate(
  "Chest Pain Character",
  .after = "cohort_unique_id"
) |>
pointblank::col_vals_in_set(
  columns = c("chest_pain_type"),
  set = c("0", "1", "2", "3", "4")
)

```

```

if (params$show_table) {
  chest_pain_type_data |>
    dplyr::distinct(.data[["Dyspnea"]], .data[["Chest Pain Character"]],
      .keep_all = TRUE) |>
    knitr::kable()
}

```

cohort_unique_id	Chest Pain Character	Dyspnea	chest_pain_type
B001	atypical	no	2
B002	no chest pain	yes	4
B003	atypical	yes	2
B005	typical	no	1
B006	nonanginal	yes	3
B007	no chest pain	no	0
B013	nonanginal	no	3
B015	typical	yes	1

Remove unnecessary columns so that we can merge with the other fields.

```

chest_pain_type_data <- chest_pain_type_data |>
  dplyr::select(-c("Dyspnea", "Chest Pain Character"))

```

3.2.4.5 Combined chest pain related tables

We combine all chest related tables together

```

join_specification <- dplyr::join_by("cohort_unique_id")

chest_pain_data <- cohort_B_data |>
  dplyr::select(c("cohort_unique_id")) |>
  dplyr::inner_join(have_chest_pain_data,
                    by = join_specification,
                    unmatched = "error",
                    relationship = "one-to-one") |>
  dplyr::inner_join(chest_pain_type_data,
                    by = join_specification,
                    unmatched = "error",
                    relationship = "one-to-one") |>
  dplyr::inner_join(shortness_of_breath_data,
                    by = join_specification,
                    unmatched = "error",
                    relationship = "one-to-one") |>
  dplyr::inner_join(symptoms_data,
                    by = join_specification,
                    unmatched = "error",
                    relationship = "one-to-one")

testthat::expect_true(
  pointblank::has_columns(
    chest_pain_data,
    columns = c("have_sob", "have_chest_pain", "symptoms",
               ↵ "chest_pain_type")
  )
)

testthat::expect_equal(
  ncol(chest_pain_data), 5
)

```

3.2.5 Combine Demographics

We combine all the data to give the `demo_behave_data`.

```

join_specification <- dplyr::join_by("cohort_unique_id")

demo_behave_data <- cohort_B_data |>
  dplyr::select(c("cohort_unique_id")) |>
  dplyr::inner_join(age_gender_data,
                    by = join_specification,
                    unmatched = "error",

```

```

        relationship = "one-to-one") |>
dplyr::inner_join(body_measurement_data,
  by = join_specification,
  unmatched = "error",
  relationship = "one-to-one") |>
dplyr::inner_join(smoking_data,
  by = join_specification,
  unmatched = "error",
  relationship = "one-to-one") |>
dplyr::inner_join(chest_pain_data,
  by = join_specification,
  unmatched = "error",
  relationship = "one-to-one") |>
dplyr::relocate(c("bsa_m2", "bmi"),
  .after = "sex")

testthat::expect_true(
  pointblank::has_columns(
    demo_behave_data,
    columns = c(
      "age_years", "sex",
      "height_cm", "weight_kg", "bsa_m2", "bmi",
      "smoke_current", "smoke_past",
      "have_sob", "have_chest_pain",
      "symptoms", "chest_pain_type"
    )
  )
)

testthat::expect_equal(
  ncol(demo_behave_data), 13
)

```

3.3 Write Preprocessed File

We output data to be used for the next session.

```

demo_behave_data |>
fst::write_fst(
  path = here::here(
    params$analysis_folder,
    params$harmonisation_folder,
    params$preprocessing_folder,

```

```
"02_demographic_data.fst"),  
)
```

4 Export To Excel

```
out_type <- knitr::opts_chunk$get("rmarkdown.pandoc.to")
```

4.1 Read all tabular data

We read all tabular data from the previous section.

4.2 Export Data as Excel

We export the standardised data to an excel file called harmonised__Cohort__B.xlsx

```
# Create a new workbook
my_workbook <- openxlsx::createWorkbook()

sheet_name = c("demographics")

output_data = list(demo_behave_data) |>
  purrr::map(
    .f = harmonisation::add_cohort_name,
    cohort_name = params$cohort_name,
    cohort_name_column = "cohort_name"
  )

purrr::walk2(
  .x = sheet_name,
  .y = output_data,
  .f = harmonisation::write_to_sheet,
  workbook = my_workbook
)

# Save workbook
openxlsx::saveWorkbook(
  wb = my_workbook,
  file = here::here(params$analysis_folder,
    params$output_folder,
```

```
        params$cleaned_folder,  
        params$output_excel_file),  
  overwrite = TRUE  
)
```